

CHAPTER 5

Morphology of Flowering Plants

The Root

- The roots that originate from the base of the stem are: (2020)
 - Primary roots
 - Prop roots
 - Lateral roots
 - Fibrous roots
- Pneumatophores occur in (2018)
 - Halophytes
 - Free-floating hydrophytes
 - Carnivorous plants
 - Submerged hydrophytes
- Sweet potato is a modified (2018)
 - Stem
 - Adventitious root
 - Tap root
 - Rhizome
- Root hairs develop from the region of: (2017-Delhi)
 - Maturation
 - Elongation
 - Root cap
 - Meristematic activity
- Roots play insignificant role in absorption of water in: (2015 Re)
 - Pistia*
 - Pea
 - Wheat
 - Sunflower

The Stem

- Which of the following statements is not correct? (2022 Re)
 - The rhizome is thick, prostrate and branched
 - Rhizome is a condensed form of stem
 - The apical bud in rhizome always remains above the ground
 - The rhizome is aerial with no distinct nodes and internodes
- In *Bougainvillea*, thorns are the modifications of: (2017-Delhi)
 - Stipules
 - Adventitious root
 - Stem
 - Leaf
- Stems modified into flat green organs performing the functions of leaves are known as: (2016 - I)
 - Cladodes
 - Phyllodes
 - Phylloclades
 - Scales
- Which of the following is not a stem modification? (2016 - I)
 - Pitcher of *Nepenthes*
 - Thorns of *Citrus*
 - Tendrils of cucumber
 - Flattened structures of *Opuntia*

- An example of edible underground stem is: (2014)
 - Potato
 - Carrot
 - Groundnut
 - Sweet potato

The Leaf

- Identify the correct set of statements: (2022)
 - The leaflets are modified into pointed hard thorns in *Citrus* and *Bougainvillea*
 - Axillary buds form slender and spirally coiled tendrils in cucumber and pumpkin
 - Stem is flattened and fleshy in *Opuntia* and modified to perform the function of leaves
 - Rhizophora* shows vertically upward growing roots that help to get oxygen for respiration
 - Subaerially growing stems in grasses and strawberry help in vegetative propagation

Choose the correct answer from the options given below.

 - A, B, D and E only
 - B and C only
 - A and D only
 - B, C, D and E only
- Leaves become modified into spines in: (2015)
 - Onion
 - Silk Cotton
 - Opuntia*
 - Pea

The Inflorescence

- Given below are two statements: One is labelled as Assertion A and the other is labeled as Reason R: (2023)

Assertion A: A flower is defined as modified shoot wherein the shoot apical meristem changes to floral meristem.

Reason R: Internode of the shoot gets condensed to produce different floral appendages laterally at successive nodes instead of leaves.

In the light of the above statements, choose the correct answer from the options given below:

 - A is false but R is true.
 - Both A and R are true and R is the correct explanation of A.
 - Both A and R are true but R is NOT the correct explanation of A.
 - A is true but R is false.

The Flower

14. Match List-I with List-II: (2024 Re)

List-I		List-II	
(A)	Vexillary aestivation	(I)	Brinjal
(B)	Epipetalous stamens	(II)	Peach
(C)	Epiphyllous stamens	(III)	Pea
(D)	Perigynous flower	(IV)	Lily

Choose the **correct** answer from the option given below:

- a. A-III, B-I, C-IV, D-II b. A-III, B-IV, C-I, D-II
c. A-III, B-II, C-I, D-IV d. A-II, B-I, C-IV, D-III

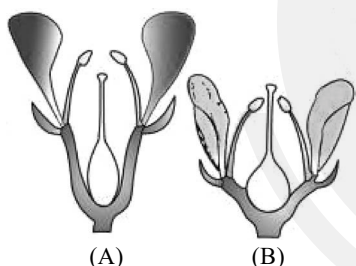
15. Match List-I with List-II: (2024 Re)

List-I		List-II	
(A)	China rose	(I)	Free central
(B)	Mustard	(II)	Basal
(C)	Primrose	(III)	Axile
(D)	Marigold	(IV)	Parietal

Choose the **correct** answer from the options given below:

- a. A-IV, B-III, C-II, D-I b. A-II, B-III, C-IV, D-I
c. A-III, B-IV, C-I, D-II d. A-III, B-IV, C-II, D-I

16. Identify the type of flowers based on the position of calyx, corolla and androecium with respect to the ovary from the given figures (A) and (B). (2024)



- a. (A) Perigynous; (B) Epigynous
b. (A) Perigynous; (B) Perigynous
c. (A) Epigynous; (B) Hypogynous
d. (A) Hypogynous; (B) Epigynous

17. Which of the following is an example of actinomorphic flower? (2024)

- a. *Pisum* b. *Sesbania* c. *Datura* d. *Cassia*

18. Match List-I with List-II. (2024)

List-I		List-II	
(A)	Rose	(I)	Twisted aestivation
(B)	Pea	(II)	Perigynous flower
(C)	Cotton	(III)	Drupe
(D)	Mango	(IV)	Marginal placentation

Choose the correct answer from the options given below:

- a. A-(IV), B-(III), C-(II), D-(I)
b. A-(II), B-(III), C-(IV), D-(I)
c. A-(II), B-(IV), C-(I), D-(III)
d. A-(I), B-(II), C-(III), D-(IV)

19. Match List-I with List-II. (2024)

List-I (Types of stamen)		List-II (Example)	
(A)	Monoadelphous	(I)	Citrus
(B)	Diadelphous	(II)	Pea
(C)	Polyadelphous	(III)	Lily
(D)	Epiphyllous	(IV)	China-rose

Choose the correct answer from the option below:

- a. A-(I), B-(II), C-(IV), B-(III)
b. A-(III), B-(I), C-(IV), D-(II)
c. A-(IV), B-(II), C-(I), D-(III)
d. A-(IV), B-(I), C-(II), D-(III)

20. Axile placentation is observed in (2023)

- a. China rose, Petunia and Lemon
b. Mustard, Cucumber and Primrose
c. China rose, Beans and Lupin
d. Tomato, Dianthus and Pea

21. In *Calotropis*, aestivation is: (2023-Manipur)

- a. Valvate b. Vexillary c. Imbricate d. Twisted

22. In a pea flower, five petals are arranged in a specialized manner with one posterior, two lateral and two anterior. These are named as and respectively. (2023-Manipur)

- a. Keel, Wings and Standard
b. Vexillum, Keel and Standard
c. Keel, Standard and Carina
d. Standard, Wings and Keel

23. Match the following: (2023-Manipur)

Type of flower		Example	
(A)	Zygomorphic	(I)	Mustard
(B)	Hypogynous	(II)	Plum
(C)	Perigynous	(III)	<i>Cassia</i>
(D)	Epigynous	(IV)	Cucumber

Select the **correct** option:

- a. A-I, B-II, C-IV, D-III b. A-I, B-II, C-III, D-IV
c. A-IV, B-I, C-III, D-II d. A-III, B-I, C-II, D-IV

24. Match List-I with List-II: (2022 Re)

List-I		List-II	
(A)	Imbricate	(I)	<i>Calotropis</i>
(B)	Valvate	(II)	<i>Cassia</i>
(C)	Vexillary	(III)	Cotton
(D)	Twisted	(IV)	Bean

Choose the correct answer from the options given below

- a. A-I, B-III, C-IV, D-II b. A-II, B-I, C-III, D-IV
c. A-II, B-I, C-IV, D-III d. A-II, B-IV, C-III, D-I

25. Which one of the following plants shows vexillary aestivation and diadelphous stamens? (2022)

- a. *Solanum nigrum* b. *Colchicum autumnale*
c. *Pisum sativum* d. *Allium cepa*

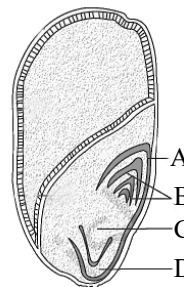
26. The flowers are Zygomorphic in: (2022)
A. Mustard B. Gulmohar C. *Cassia*
D. *Datura* E. Chilli
Choose the correct answer from the options given below.
a. C, D and E only b. A, B and C only
c. B and C only d. D and E only
27. Diadelphous stamens are found in: (2021)
a. Citrus b. Pea
c. China rose and citrus d. China rose
28. Ray florets have: (2020)
a. Superior ovary b. Hypogynous ovary
c. Half inferior ovary d. Inferior ovary
29. The ovary is half inferior in: (2020)
a. Mustard b. Sunflower
c. Plum d. Brinjal
30. Correct position of floral parts over thalamus in mustard plant is- (2020-Covid)
a. Margin of the thalamus grows upward, enclosing the ovary completely, and other parts arise below the ovary
b. Gynoecium is present in the centre and other parts cover it partially
c. Gynoecium is situated in the centre, and other parts of the flower are located at the rim of the thalamus, at the same level
d. Gynoecium occupies the highest position, while the other parts are situated below it
31. Placentation in which ovules develop on the inner wall of the ovary or in peripheral part, is (2019)
a. Basal b. Axile
c. Parietal d. Free central
32. Free-central placentation is found in: (2016 - II)
a. *Brassica* b. *Citrus* c. *Dianthus* d. *Argemone*
33. Radial symmetry is found in the flowers of: (2016 - II)
a. *Pisum* b. *Cassia* c. *Brassica* d. *Trifolium*
34. How many plants among *Indigofera*, *Sesbania*, *Salvia*, *Allium*, Aloe, mustard, groundnut, radish, gram and turnip have stamens with different lengths in their flowers? (2016 - II)
a. Five b. Six c. Three d. Four
35. The term 'Polyadelphous' is related to: (2016 - II)
a. Corolla b. Calyx
c. Gynoecium d. Androecium
36. The standard petal of a papilionaceous corolla is also called: (2016 - I)
a. Carina b. Pappus c. Vexillum d. Corona
37. Perigynous flowers are found in: (2015)
a. China rose b. Rose
c. Guava d. Cucumber
38. Ovary is inferior in: (2015)
a. *Guava* b. *Rose* c. *China rose* d. *Peach*
39. Axile placentation is present in: (2015 Re)
a. Lemon b. Pea c. *Argemone* d. *Dianthus*
40. Among china rose, mustard, brinjal, potato, onion and tulip, how many plants have superior ovary? (2015 Re)
a. Six b. Three c. Four d. Five
41. When the margins of sepals or petals overlap one another without any particular direction, the condition is termed as: (2014)
a. Valvate b. Vexillary c. Imbricate d. Twisted
42. Among bitter gourd, mustard, brinjal, pumpkin, china rose, *Lupin*, cucumber, sun hemp, gram, guava, bean, chili, plum, *Petunia*, tomato, rose, *Withania somnifera*, potato, onion, Aloe and tulip how many plants have hypogynous flower? (2013)
a. Eighteen b. Six c. Ten d. Fifteen
43. In China rose the flowers are: (2013)
a. Zygomorphic, epigynous with twisted aestivation
b. Actinomorphic, hypogynous with twisted aestivation
c. Actinomorphic, epigynous with valvate aestivation
d. Zygomorphic, hypogynous with imbricate aestivation

The Fruit

44. Identify the correct features of Mango and Coconut fruits.
(i) In both fruit is a drupe
(ii) Endocarp is edible in both
(iii) Mesocarp in Coconut is fibrous, and in Mango it is fleshy
(iv) In both, fruit develops from monocarpellary ovary
Select the correct option from below: (2020-Covid)
a. (i), (ii) and (iii) only b. (i) and (iv) only
c. (i) and (ii) only d. (i), (iii) and (iv) only
45. Coconut fruit is a: (2017-Delhi)
a. Drupe b. Berry c. Nut d. Capsule
46. The morphological nature of the edible part of coconut is: (2017-Delhi)
a. Perisperm b. Cotyledon
c. Endosperm d. Pericarp
47. An aggregate fruit is one which develops from: (2014)
a. Multicarpellary superior ovary
b. Multicarpellary syncarpous gynoecium
c. Multicarpellary apocarpous gynoecium
d. Complete inflorescence
48. Placenta and pericarp are both edible portions in: (2014)
a. Potato b. Apple c. Banana d. Tomato

The Seed

49. Identify the part of the seed from the given figure which is destined to form root when the seed germinates. (2024)

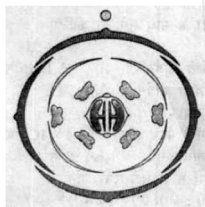


- a. C b. D c. A d. B

50. Which one of the following statements is correct? (2014)
 a. A sterile pistil is called a staminode
 b. The seed in grasses is not endospermic
 c. Mango is a parthenocarpic fruit
 d. A proteinaceous aleurone layer is present in maize grain
51. Seed coat is not thin, membranous in: (2013)
 a. Gram b. Maize c. Coconut d. Groundnut

Description of Families-Fabaceae, Solanaceae, Liliaceae

52. Which of the following examples show monocarpellary, unilocular ovary with many ovules? (2024 Re)
 A. *Sesbania* B. Brinjal C. *Indigofera*
 D. Tobacco E. *Asparagus*
- Choose the **correct** answer from the options given below:
 a. B and E only b. C, D and E only
 c. A, B and D only d. A and C only
53. Family Fabaceae differs from Solanaceae and Liliaceae. With respect to the stamens, pick out the characteristics specific to family Fabaceae but not found in Solanaceae or Liliaceae. (2023)
 a. Epiphyllous and Ditheous anthers
 b. Diadelphous and Ditheous anthers
 c. Polyadelphous and epipetalous stamens
 d. Monoadelphous and Monotheous anthers
54. The Floral Diagram represents which one of the following families (2022 Re)



- a. Liliaceae b. Fabaceae
 c. Brassicaceae d. Solanaceae

55. Match Column - I with Column - II (2021)

	Column-I		Column-II
A.	$\% \text{O} \nearrow \text{K}_{(5)} \text{C}_{1+2+(2)} \text{A}_{(9)+1} \underline{\text{G}}_1$	(i)	Brassicaceae
B.	$\oplus \text{O} \nearrow \text{K}_{(5)} \text{C}_{(5)} \text{A}_5 \underline{\text{G}}_2$	(ii)	Liliaceae
C.	$\oplus \text{O} \nearrow \text{P}_{(3+3)} \text{A}_{3+3} \underline{\text{G}}_{(3)}$	(iii)	Fabaceae
D.	$\oplus \text{O} \nearrow \text{K}_{2+2} \text{C}_4 \text{A}_{2-4} \underline{\text{G}}_{(2)}$	(iv)	Solanaceae

Select the correct answer from the options given below.

- a. A-i B-ii C-iii D-iv
 b. A-ii B-iii C-iv D-i
 c. A-iv B-ii C-i D-iii
 d. A-iii B-iv C-ii D-i
56. Which of the following is the correct floral formula of Liliaceae? (2020-Covid)
 a. $\oplus \text{O} \nearrow \text{K}_{(5)} \text{C}_{(5)} \text{A}_5 \underline{\text{G}}_{(2)}$ b. $\text{Br} \oplus \text{O} \nearrow \text{P}_{(3+3)} \text{A}_{3+3} \underline{\text{G}}_{(3)}$
 c. $\oplus \text{O} \nearrow \text{K}_{(5)} \text{C}_{(5)} \text{A}_5 \underline{\text{G}}_{(2)}$ d. $\% \text{O} \nearrow \text{C}_{1+2+(2)} \text{A}_{(9)+1} \underline{\text{G}}_{(1)}$
57. Tricarpellary, syncarpous gynoecium is found in flowers of: (2016 - I)
 a. Liliaceae b. Solanaceae
 c. Fabaceae d. Poaceae
58. Keel is the characteristic feature of flower of: (2015)
 a. Aloe b. Tomato
 c. Tulip d. *Indigofera*

Answer Key

1. (d) 2. (a) 3. (b) 4. (a) 5. (a) 6. (d) 7. (c) 8. (c) 9. (a) 10. (a)
 11. (d) 12. (c) 13. (b) 14. (a) 15. (c) 16. (b) 17. (c) 18. (c) 19. (c) 20. (a)
 21. (a) 22. (d) 23. (d) 24. (c) 25. (c) 26. (c) 27. (b) 28. (d) 29. (c) 30. (d)
 31. (c) 32. (c) 33. (c) 34. (d) 35. (d) 36. (c) 37. (b) 38. (a) 39. (a) 40. (a)
 41. (c) 42. (d) 43. (b) 44. (d) 45. (a) 46. (c) 47. (c) 48. (d) 49. (a) 50. (d)
 51. (c) 52. (d) 53. (b) 54. (c) 55. (d) 56. (b) 57. (a) 58. (d)